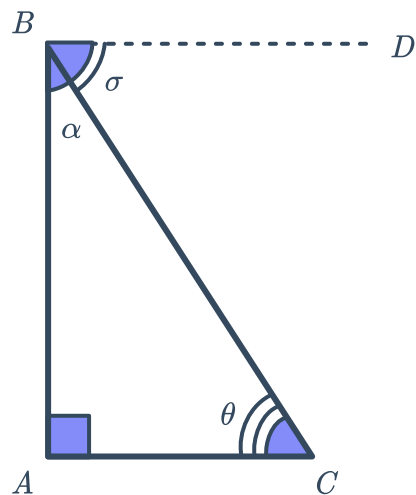


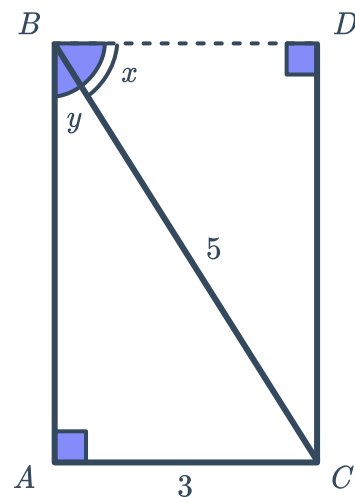
Angles of elevation and depression



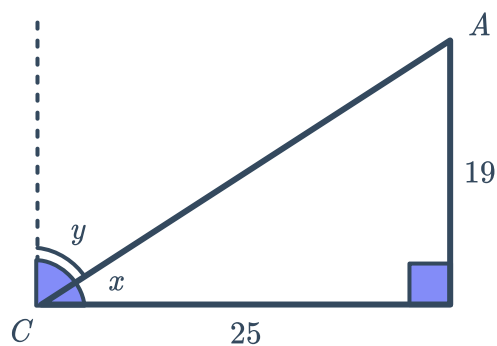
- 1 A man standing at the top of the tower at point B , is looking at the ground at point C . Identify the angle of depression in the following figure:



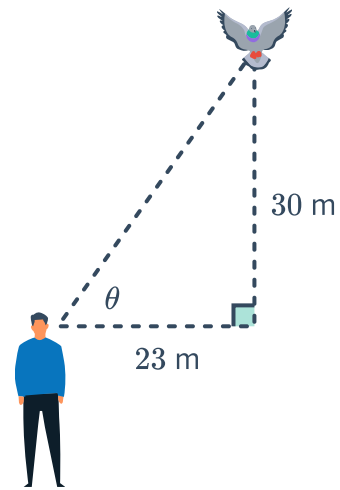
- 2 Find the angle of depression from point B to point C in the diagram shown. Round your answer to two decimal places.



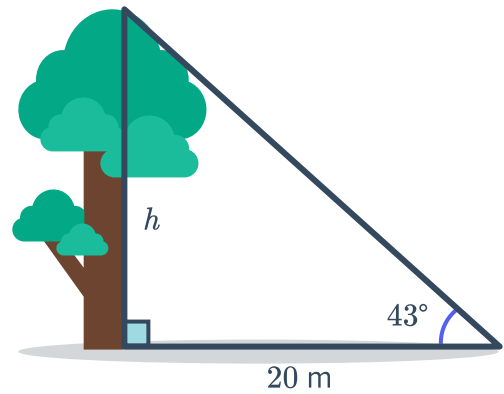
- 3 Find x , the angle of elevation to point A from point C . Round your answer to two decimal places.



- 4 The person in the picture sights a pigeon above him. Find the angle of elevation, θ , to two decimal places.

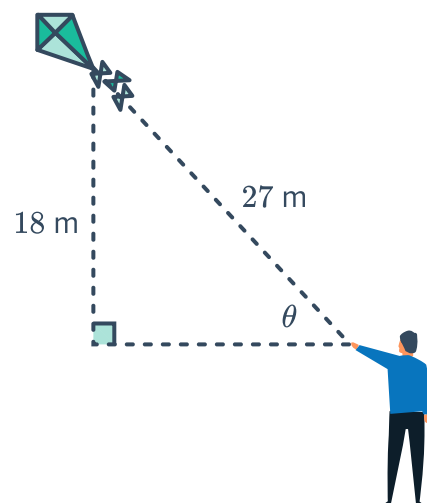


- 5 Sally measures the angle of elevation to the top of a tree from a point 20 m away to be 43° . Find the height of the tree, h , to the nearest whole number.

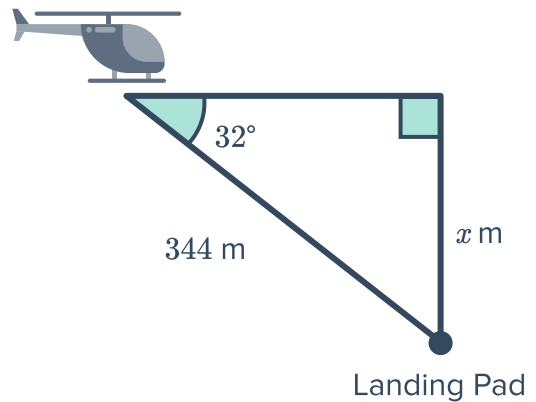


- 6 A boy flying his kite releases the entire length of his string which measures 27 m, so that the kite is 18 m above him.

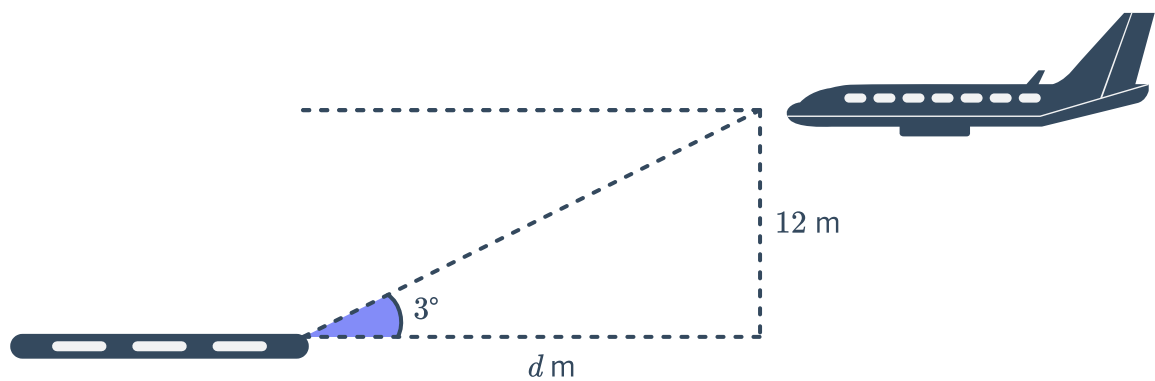
If the angle the string makes with the horizontal ground is θ , find θ to two decimal places.



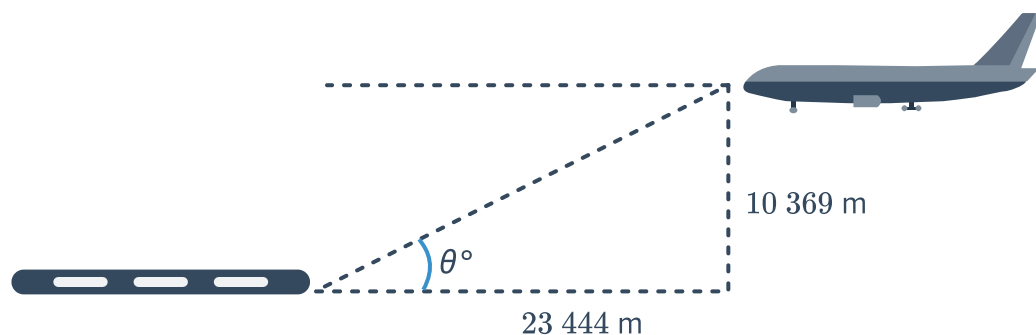
- 7 A helicopter is 344 m away from its landing pad. If the angle of depression to the landing pad is 32° , find x , the height of the helicopter above the ground, to the nearest metre.



- 8 The final approach of an aeroplane when landing requires the pilot to adjust the angle of descent to about 3° as shown in the diagram below. If the plane is 12 m above the runway and has d m until touchdown, find d to the nearest metre.



- 9 The airtraffic controller is communicating with a plane in flight approaching an airport for landing. The plane is 10 369 m above the ground and is still 23 444 m from the runway. If θ° is the angle at which the plane should approach, find θ to one decimal place.



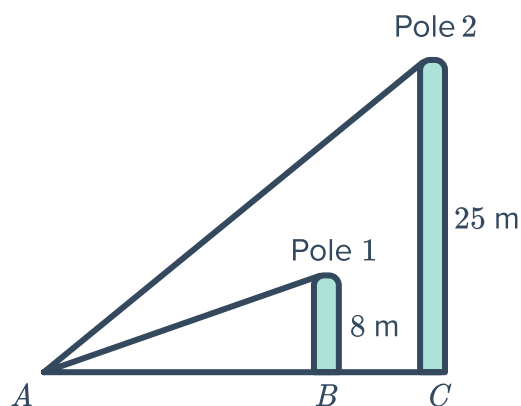


Applications

- 10** Fred is on a ship and observes a lighthouse on a cliff in the distance. The base of the cliff is 651 m away from the ship, and the angle of elevation of the top of the lighthouse from Fred is 35° .
- a** If the top of the lighthouse is x m above sea level, find x correct to two decimal places.
 - b** If the lighthouse is 30 metres tall, how tall is the cliff that the lighthouse stands on? Round your answer correct to two decimal places.
- 11** Buzz is standing 250 m from a building and measures the angle of elevation of the top of the building to be 27° .
- a** If the difference in height between the top of the building and Buzz's eye is h m, find h correct to two decimal places.
 - b** If Buzz's eye is 172 cm from the ground, find the height of the building correct to one decimal place.
- 12** From the top of a rocky ledge that is 274 m high, the angle of depression to a boat is 15° . If the boat is d m from the foot of the cliff, find d correct to two decimal places.
- 13** Amelia measures the angle of elevation to the top of a tree from a point, 29 m away from the base, to be 31° . Find the height of the tree, h , to the nearest metre.
- 14** The angle of elevation from an observer to the top of a tree is 18° . If the distance between the tree and the observer is d m and the tree is known to be 3.53 m high, find the value of d to two decimal places.
- 15** At a certain time of the day a light post, 6 m tall, has a shadow of 5.8 m. If the angle of elevation of the sun at that time is θ° , find θ to two decimal places.

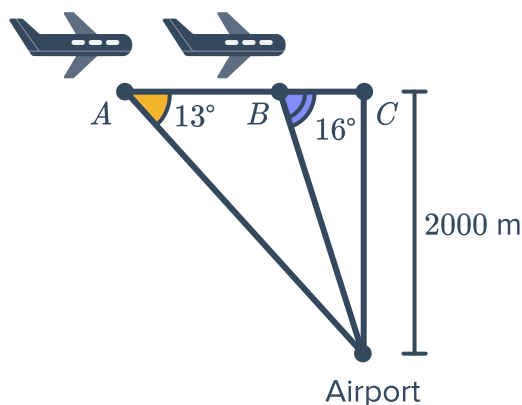
- 16 A man stands at point A looking at the top of two poles. Pole 1 has a height 8 m and an angle of elevation of 34° from point A . Pole 2 has a height 25 m and an angle of elevation of 57° from point A .

- a Find the distance from A to B , to two decimal places.
- b Find the distance from A to C , to two decimal places.
- c Hence, find BC , the distance between the two poles in metres. Round your answer to one decimal place.



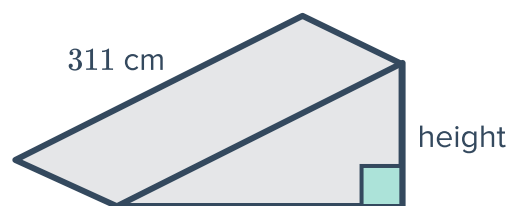
- 17 A fighter jet, flying at an altitude of 2000 m is approaching an airport. The pilot measures the angle of depression to the airport to be 13° . One minute later, the pilot measures the angle of depression again and finds it to be 16° .

- a Find the distance AC , to the nearest metre.
- b Find the distance BC , to the nearest metre.
- c Hence, find the distance covered by the jet in that one minute, to the nearest metre.



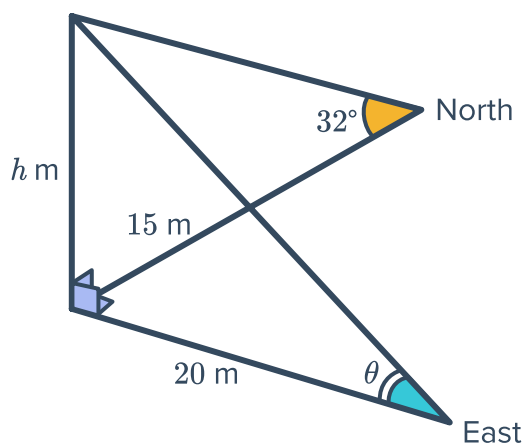
18 A ramp of length 311 cm needs to ascend  angle between 10° and 20° for it to be safe to use.

- a If the height of the ramp is 152 cm, and the angle the ramp makes with the ground is x , find x to two decimal places.
- b If the height of the ramp is 25 cm, and the angle the ramp makes with the ground is y , find y to two decimal places.
- c If the height of the ramp is 100 cm, and the angle the ramp makes with the ground is z , find z to two decimal places.
- d Hence, at which height is the ramp safe?



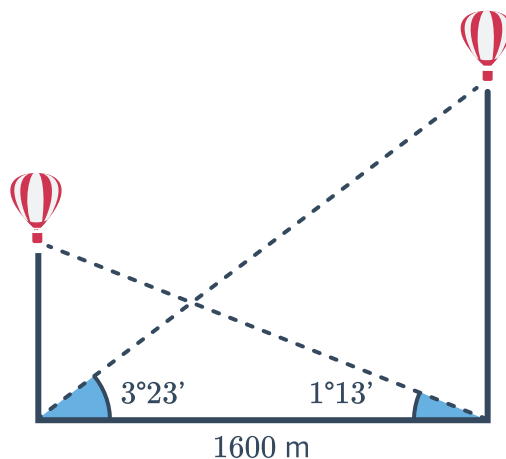
19 From a point 15 m due north of a tower, the angle of elevation of the tower is 32° .

- a Find the height of the tower h . Round your answer to two decimal places.
- b Find the size θ of the angle of elevation of the tower at a point 20 m due east of the tower. Round your answer to the nearest degree.



20 Two hot air balloons are tied to the ground  , each at a different location. An observer at each location measures the angle of elevation to the opposite balloon. The observers are 1600 m apart as shown in the diagram:

- a Calculate the difference in height between the two balloons. Round your answer to the nearest metre.
- b Hence, find the direct distance between the two balloons. Round your answer to the nearest metre.



21 Roald is standing at point P and observes two poles, AB and CD , of different heights. P , B , and D are on horizontal ground:

From P , the angles of elevation to the top of the poles at A and C are 29° and 18° respectively. Roald is 16 m from the base of pole AB . The height of pole CD is 7 m.

- a Calculate the distance from Roald to the top of pole CD , to two decimal places.
- b Calculate the distance from Roald to the top of pole AB , to two decimal places.

